

## Methods and goals in AI

### Symbolic vs. connectionist approaches

AI research follows two distinct, and to some extent competing, methods, the symbolic (or “top-down”) approach, and the connectionist (or “bottom-up”) approach. The top-down approach seeks to replicate intelligence by analyzing cognition independent of the biological structure of the brain, in terms of the processing of symbols—whence the symbolic label. The bottom-up approach, on the other hand, involves creating artificial neural networks in imitation of the brain’s structure—whence the connectionist label.

To illustrate the difference between these approaches, consider the task of building a system, equipped with an optical scanner, that recognizes the letters of the alphabet. A bottom-up approach typically involves training an artificial neural network by presenting letters to it one by one, gradually improving performance by “tuning” the network. (Tuning adjusts the responsiveness of different neural pathways to different stimuli.) In contrast, a top-down approach typically involves writing a computer program that compares each letter with geometric descriptions. Simply put, neural activities are the basis of the bottom-up approach, while symbolic descriptions are the basis of the top-down approach.

In *The Fundamentals of Learning* (1932), Edward Thorndike, a psychologist at Columbia University, New York City, first suggested that human learning consists of some unknown property of connections between neurons in the brain. In *The Organization of Behavior* (1949), Donald Hebb, a psychologist at McGill University, Montreal, suggested that learning specifically involves strengthening certain patterns of neural activity by increasing the probability (weight) of induced neuron firing between the associated connections.

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In 1957 two vigorous advocates of symbolic AI—Allen Newell, a researcher at the RAND Corporation, Santa Monica, California, and Herbert Simon, a psychologist and computer scientist at Carnegie Mellon University, Pittsburgh—summed up the top-down approach in what they called the physical symbol system hypothesis. This hypothesis states that processing structures of symbols is sufficient, in principle, to produce artificial intelligence in a digital computer and that, moreover, human intelligence is the result of the same type of symbolic manipulations.

During the 1950s and '60s the top-down and bottom-up approaches were pursued simultaneously, and both achieved noteworthy, if limited, results. During the 1970s, however, bottom-up AI was neglected, and it was not until the 1980s that this approach again became prominent. Nowadays both approaches are followed, and both are acknowledged as facing difficulties. Symbolic techniques work in simplified realms but typically break down when confronted with the real world; meanwhile, bottom-up researchers have been unable to replicate the nervous systems of even the simplest living things. *Caenorhabditis elegans*, a much-studied worm, has approximately 300 neurons whose pattern of interconnections is perfectly known. Yet connectionist models have failed to mimic even this worm. Evidently, the neurons of connectionist theory are gross oversimplifications of the real thing.